



Engineering Design & Mechanical Analysis of an 8-DOF Quadraped Robot

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1. Project Overview

This report covers the comprehensive mechanical analysis and structural design of an 8-DOF Quadruped Robot. The system simulates organic mammal locomotion using a dual-link leg mechanism.

2. Chassis Geometric Design

The robot base is designed as a solid rectangular chassis with specific dimensions of (120 mm × 70 mm × 15 mm).

Mechanical Calculations

Weight and Force Analysis (F_g)

$$F_g = m \times g$$

Where :

$$m = 0.1 \text{ Kg}$$

$$g = 9.8 \text{ m/s}^2$$

$$F = 1.0 \text{ Kg} \times 9.8 \text{ m/s}^2 = 9.81$$

Joint Torque (T)

$$F_{\text{leg}} = \frac{F_g}{4} = \frac{9.81 \text{ N}}{4} = 2.4525 \text{ N}$$

$$T = F_{\text{leg}} \times L$$

$$T = 2.4525 \times 0.045 \text{ m} = 0.1103 \text{ N.m}$$

$$= 0.1103 \times 10.197 \approx 1.125 \text{ Kg.cm}$$

Factor of safety (Fos)

$$Fos = \frac{1.8}{1.125} = 1.6$$

Degrees of freedom (DoF)

$$DoF_{\text{total}} = 4 \text{ Legs} \times 2 \text{ DoF/Leg} = 8 \text{ DoF}$$